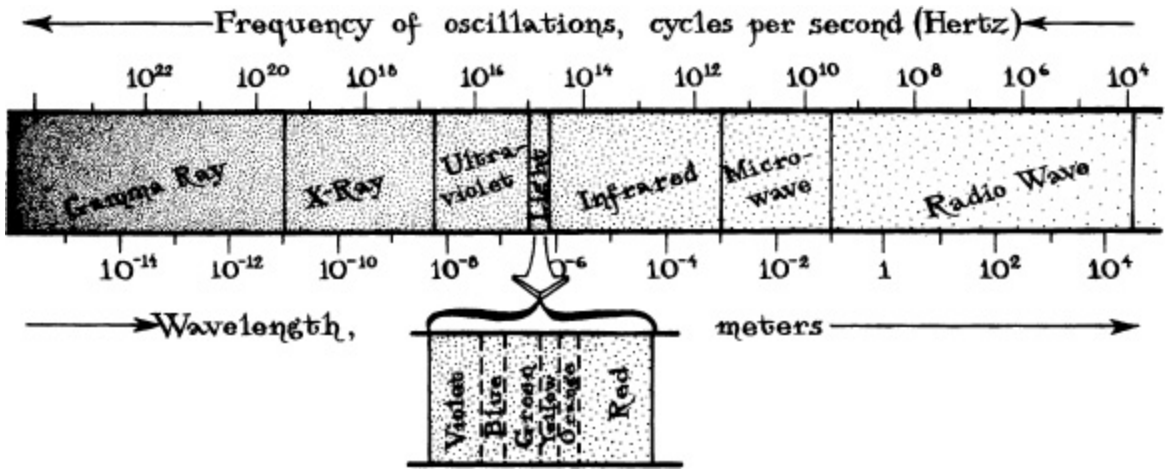


P.2 The spectrum of electromagnetic waves, running from radio waves at very long wavelengths (very low frequencies) to gamma rays at very short wavelengths (very high frequencies). For a discussion of the notation used here for numbers (10^{21}, 10^{-12}, etc.), see Box P.1 below.

At first you explore passively: You use instrumented telescopes to study the electromagnetic waves (the radiation) that the gas emits as it streams toward the hole. Far from the hole, the gas atoms are cool, just a few degrees above absolute zero. Being cool, they vibrate slowly; and their slow vibrations produce slowly oscillating electromagnetic waves, which means waves with long distances from one crest to the next—long wavelengths. These are radio waves; see Figure P.2. Nearer the hole, where gravity has pulled the atoms into a faster stream, they collide with each other and heat up to several thousand degrees. The heat makes them vibrate more rapidly and emit more rapidly oscillating, shorter wavelength waves, waves that you recognize as light of varied hues: red, orange, yellow, green, blue, violet (Figure P.2). Much closer to the hole, where gravity is much stronger and the stream much faster, collisions heat the atoms to several million degrees, and they vibrate very fast, producing electromagnetic waves of very short wavelength: X-rays. Seeing those X-rays pour out of the hole's vicinity, you are reminded that it was by discovering and studying just such X-rays that astrophysicists, in 1972, identified the first black hole in distant space: Cygnus X-1, 6,000 light-years from Earth.[3]

Turning your telescopes still closer to the hole, you see gamma rays from atoms heated to still higher temperatures. Then, looming up, at the center of this brilliant display, you see a large, round sphere, absolutely black; it is the black hole, blotting out all the light, X-rays, and gamma rays